

Semantic Drift in the Qwen3.5 Model Family: Scaling, Quantization, Temperature, and Prompt Engineering

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Abstract

We extend the semantic drift framework of Smith [2026] to the Qwen3.5 model family, conducting a controlled within-family analysis across 18 GGUF models (0.8B–122B parameters, 7 sizes and 18 size/quant combinations), an AWQ and a GPTQ-Int4 variant of the 35B MoE served via vLLM, an NVFP4 variant served via sglang, a 9B AWQ variant, and an out-of-family Nemotron-3-Nano-30B NVFP4 control. Across 1,969 completed 30-iteration chains comprising 59,070 LLM inferences, we investigate four research questions: **RQ1** how drift varies with model size and quantization, **RQ2** whether GGUF, AWQ, NVFP4, and GPTQ-Int4 quantization produce different drift dynamics for the same 35B MoE architecture, **RQ3** how sampling temperature interacts with model size, and **RQ4** whether system prompt effects generalize across model sizes and quantization formats. Key findings include: (1) a stability threshold at approximately 4B parameters, below which drift becomes unpredictable; (2) quantization effects that are non-monotonic and size-dependent—BF16 does not always outperform Q4_K_M; (3) clear differences between 4-bit quantization *methods* for the same 35B MoE architecture, although the per-method confidence intervals overlap at $n \leq 5$ chains per cell; (4) inverted temperature sensitivity in smaller models. A cross-generation comparison with Qwen3 at matched parameter counts is also reported, with the caveat that the Qwen3 arm comes from single unreplicated pilot runs.

1 Introduction

In Smith [2026] (hereafter “Paper 1”), we established a telephone-game framework for measuring semantic drift in iterated LLM paraphrase chains. That study examined 17 diverse models, KV cache precision effects, system prompts, and temperature, finding a stable four-tier model hierarchy, non-monotonic infrastructure effects, and a complexity threshold for prompt engineering.

Paper 1 compared models *across* families—Qwen3, Gemma3, Llama 3.1, Devstral, GLM, and others—making it difficult to disentangle model-family effects from size and quantization effects. The present study addresses this limitation by conducting a controlled *within-family* analysis of the Qwen3.5 model family, holding architecture constant while varying size (0.8B–122B), quantization (BF16, Q8_0, Q4_K_M), serving backend (Ollama GGUF vs. vLLM AWQ), temperature, and system prompts.

Our contributions are:

1. A systematic scaling analysis across 7 Qwen3.5 model sizes at 3 quantization levels, revealing a stability threshold and non-monotonic quantization effects (Section 4.1).

2. A controlled four-way comparison of GGUF, AWQ, NVFP4, and GPTQ-Int4 quantization for the same 35B MoE architecture (Section 4.2).
3. A temperature ablation across 5 model sizes showing inverted sensitivity in small models (Section 4.3).
4. A system prompt ablation across six Qwen3.5 configurations and one out-of-family control, testing whether Paper 1’s prompt engineering findings generalize within a single model family (Section 4.4).
5. A cross-generation comparison between Qwen3 and Qwen3.5 at matched parameter counts (Section 4.6).

2 Related Work

We refer readers to Paper 1 [Smith, 2026] for a comprehensive review of semantic drift, cultural attractors [Perez et al., 2025, Brinkmann et al., 2023], iterative generation distortion [Mohamed et al., 2025], agent drift [Rath, 2026], attractor cycles [Wang et al., 2025, Chytas and Singh, 2025], and quantization effects [Han et al., 2026, Hooper et al., 2024, Liu et al., 2024].

The present study extends this literature by providing the first controlled within-family scaling analysis of semantic drift. While Paper 1 demonstrated that model choice dominates infrastructure and prompt engineering, it left open whether this hierarchy holds when architecture is controlled. The Qwen3.5 family is well-suited for this analysis: it spans two orders of magnitude in parameter count (0.8B–122B), includes both dense and mixture-of-experts architectures, and is available in multiple quantization formats across serving backends.

3 Methodology

3.1 Experimental Framework

We reuse the telephone-game framework from Paper 1: a single text is paraphrased 30 times in sequence by the same model (“paraphrase” prompt style), with cosine similarity to the original measured at each iteration using Qwen3-Embedding-0.6B. All experiments use 5 random seeds {42, 137, 256, 512, 1024} unless otherwise noted. Temperature is 0.7 by default.

3.2 Prompts

We use the same two prompts as Paper 1:

- **Spaghetti** (3 steps): “Step 1: Boil water. Step 2: Add pasta for 8 minutes. Step 3: Drain and serve with sauce.”
- **Lasagna** (21 steps): A detailed homemade beef lasagna recipe with ingredients, measurements, and layering instructions (see Appendix A).

3.3 Model Inventory

Table 1 summarizes the 22 Qwen3.5 model configurations tested, together with one out-of-family control model.

Table 1: Model inventory. GGUF models were served with Ollama; the AWQ and GPTQ-Int4 variants with vLLM; the NVFP4 variants with sglang. The Nemotron-3-Nano-30B row is an out-of-family control, included to test whether the RQ4 prompt rankings are specific to the Qwen3.5 family. See Appendix E for hardware and software versions.

Size	Architecture	Quantizations	Backend
0.8B	Dense	Q8_0, BF16	Ollama
2B	Dense	Q4_K_M, Q8_0, BF16	Ollama
4B	Dense	Q4_K_M, Q8_0, BF16	Ollama
9B	Dense	Q4_K_M, Q8_0, BF16	Ollama
27B	Dense	Q4_K_M, Q8_0, BF16	Ollama
35B	MoE (A3B)	Q4_K_M, Q8_0, BF16	Ollama
35B	MoE (A3B)	AWQ 4-bit	vLLM
35B	MoE (A3B)	NVFP4 4-bit	sglang
35B	MoE (A3B)	GPTQ-Int4	vLLM
9B	Dense	AWQ 4-bit	vLLM
122B	MoE	Q4_K_M	Ollama
<i>Out-of-family control</i>			
30B	MoE (A3B)	NVFP4 4-bit	sglang

3.4 Experiments

1. **Exp 1: Size $\times$ Quantization** (18 models $\times$ 2 recipes $\times$ 5 seeds = 180 conditions).
2. **Exp 2: Temperature ablation** (4 Ollama models $\times$ 2 recipes $\times$ 5 temperatures $\times$ seeds = 168 conditions).
3. **Exp 3: vLLM AWQ (35B MoE)** — (a) baseline drift (10 conditions), (b) temperature ablation (42 conditions), (c) system prompt ablation (200 conditions).
4. **Exp 3d: sglang NVFP4 (35B MoE)** — drift baseline (10 conditions), temperature ablation (42 conditions) and system prompt ablation (200 conditions).
5. **Exp 3e: vLLM GPTQ-Int4 (35B MoE)** — drift baseline (10 conditions), temperature ablation (42 conditions) and system prompt ablation (200 conditions).
6. **Exp 3f: vLLM AWQ (9B dense)** — drift baseline (10 conditions), temperature ablation (42 conditions) and system prompt ablation (200 conditions).
7. **Exp 3g: sglang NVFP4 (Nemotron-3-Nano-30B, control)** — drift baseline (10 conditions), temperature ablation (42 conditions) and system prompt ablation (200 conditions).
8. **Exp 4: System prompt ablation** (2 Ollama models $\times$ 2 recipes $\times$ 20 prompts $\times$ 5 seeds = 400 conditions).

Counting only chains that completed all 30 iterations, the study comprises 1,969 conditions and $1,969 \times 30 = 59,070$ LLM inferences.

Data completeness. Of the 2,007 chains launched, 1,969 (98.1%) ran to iteration 30 and are the only chains used in any statistic reported here; partial chains are neither padded nor silently averaged in, and every table reports the actual n behind each cell. The shortfalls are:

- **Exp 1:** 179/180. The condition `qwen3.5:35b-q4_K_M_s256` (spaghetti) was never recorded, so the 35B MoE Q4_K_M spaghetti row rests on $n = 4$.
- **Exp 3b** (AWQ temperature): 37/42, and **Exp 3c** (AWQ system prompts): 194/200. Eleven conditions were destroyed by a failed recovery run that replaced the original chains with zero-iteration error records. The AWQ endpoint was subsequently retired and these conditions could not be regenerated.
- **Exp 4:** 378/400. Nineteen of the 22 incomplete chains are 27B lasagna conditions for prompts 15–20, lost when the Ollama serving host was decommissioned during a recovery run; the affected prompt cells therefore carry n between 0 and 4, and one cell (27B lasagna, “Balanced Clarity”) has no completed chain at all and is reported as absent rather than estimated. The remaining three are 9B lasagna chains that stopped at 15–23 iterations.
- **Exp 2 (Nemotron):** 40/42; and one system-prompt chain each for NVFP4, GPTQ-Int4 and 9B AWQ stopped between iterations 27 and 29.

3.5 Statistical Methods

Following Paper 1, we report bootstrap 95% confidence intervals ($B = 10,000$), Kruskal–Wallis H with Dunn’s post-hoc test (Bonferroni-corrected) for tier comparisons, Wilcoxon signed-rank tests for paired comparisons, Spearman rank correlations with bootstrap CIs for cross-model prompt rankings, and Cohen’s d for effect sizes.

4 Results

4.1 RQ1: Size and Quantization

Figure 1 shows drift trajectories for all 18 Ollama models on the spaghetti recipe. Table 2 provides summary statistics.

Stability threshold at 4B. Every model at or above 4B parameters reaches a mean final similarity of at least 0.82 *except* where whole chains collapse to the empty-output floor of 0.235 (27B Q4_K_M, 0.643; 35B MoE Q4_K_M, 0.579); below 4B, no configuration is reliable. The 0.8B models are the most fragile: BF16 sits at 0.235 in every chain (the embedding floor), while Q8_0 averages 0.370 with extreme seed variance (CI: 0.235–0.641). Many of these low-similarity scores result from the model returning *empty* output strings—the API returns a successful response with no content. The most likely explanation is that the model’s internal thinking process (Qwen3.5 models use a think-before-responding architecture) consumes the entire generation budget on reasoning tokens, leaving nothing for the final answer. This failure mode is consistent across seeds and becomes more frequent as chain inputs degrade through drift. The 27B models also exhibit sporadic empty outputs, particularly under Q4_K_M quantization. These results suggest a minimum capacity threshold below which models cannot reliably maintain semantic content through iterated paraphrasing.

Non-monotonic quantization. Quantization effects are not monotonic. For the 4B dense model, BF16 (0.885) leads, followed by Q4_K_M (0.861) and Q8_0 (0.824). The 2B model shows Q8_0 (0.876) far ahead of BF16 (0.636) and Q4_K_M (0.567), a reversal of more than 0.30 between the two higher-precision formats. For the 35B MoE, BF16 (0.846) and Q8_0 (0.837) are close together while Q4_K_M falls to 0.579 ($n = 4$); for the 27B dense model BF16 (0.900) and Q8_0 (0.871) likewise sit

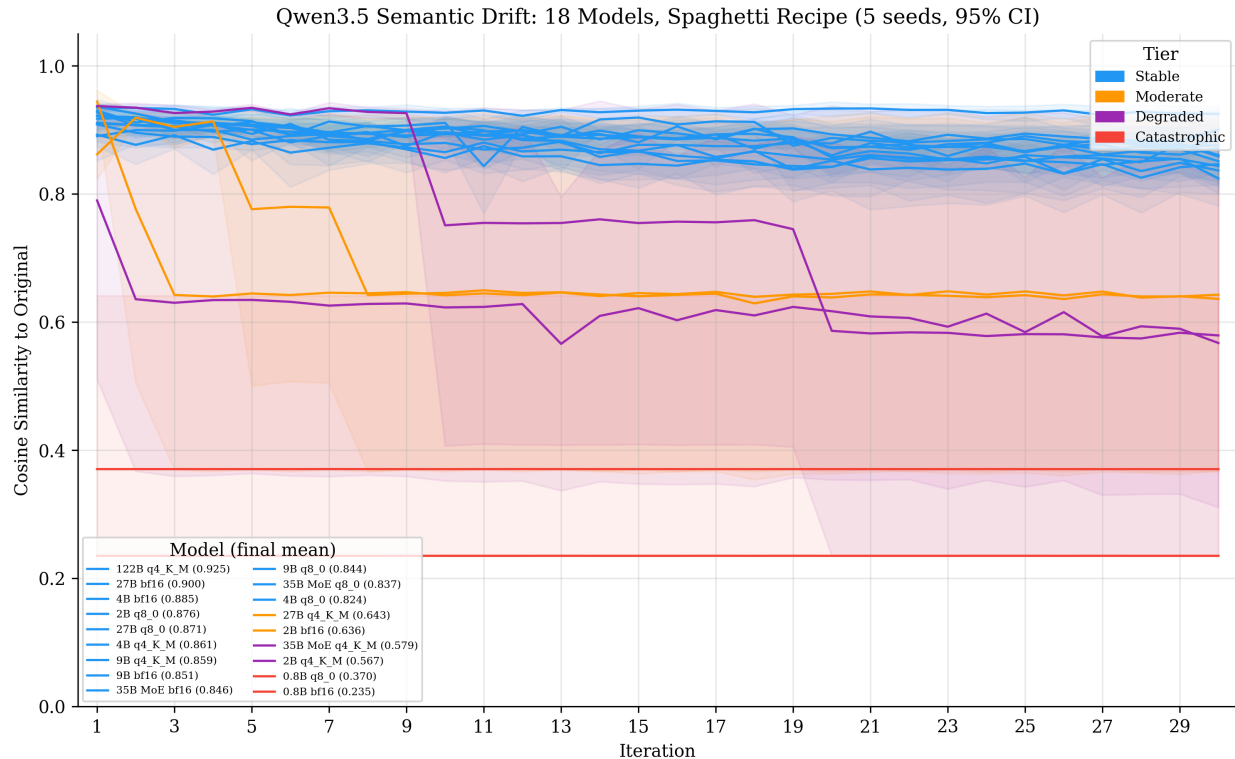


Figure 1: Semantic drift across 18 Qwen3.5 GGUF models (spaghetti recipe, up to 5 completed chains per model, 95% bootstrap CI bands). Models colored by stability tier; per-model n is given in Table 2.

well above Q4_K_M (0.643). In both of the latter cases the gap is driven by whole chains collapsing to the empty-output floor rather than by gradual degradation, and the corresponding confidence intervals are correspondingly wide (35B MoE Q4_K_M: [0.235, 0.923]). These reversals confirm Paper 1’s finding that higher precision does not guarantee less drift, but they also show that at 4-bit the dominant risk is a discrete generation failure rather than a smooth loss of fidelity.

MoE architectures. The 122B MoE achieves the family’s best score (0.925, CI [0.915, 0.937]) and is the only model whose interval excludes 0.90 from below. At comparable parameter counts the 35B MoE (0.579–0.846) does not beat the 27B dense models (0.643–0.900): on this data the sparse-activation advantage suggested by Paper 1 is not reproduced within the Qwen3.5 family once empty-output failures are counted.

Statistical tests. Kruskal–Wallis over the four occupied tiers confirms significant differences ($H = 18.19$, $p = 4.0 \times 10^{-4}$). Cohen’s d between the Stable and Catastrophic tiers is 6.18 and between Stable and Moderate 1.67. As in Paper 1, the test treats the individual seed-chains as observations even though chains from the same model are not independent of one another, so it should be read as descriptive rather than inferential.

Figure 2 presents the size $\times$ quantization heatmap.

Table 2: Qwen3.5 drift by size and quantization. Mean final cosine similarity at iteration 30 over *clean* chains only (chains with no empty, runaway-thinking generation), with 95% bootstrap CI, sorted by clean spaghetti mean descending. n_c is the number of clean 30-iteration chains behind each mean; “RA” is the runaway-thinking rate, i.e. the fraction of completed chains that contained at least one empty generation (runaway chains are excluded from the means). All-chain means are reported in the supplementary `all_stats.json`.

Size	Quant	Spaghetti				Lasagna			Tier
		Mean	95% CI	n_c	RA	Mean	n_c	RA	
122B	q4_K_M	0.925	[0.915, 0.937]	5	0.00 (0/5)	0.864	5	0.00 (0/5)	Stable
35B MoE	q4_K_M	0.923	[0.920, 0.927]	2	0.50 (2/4)	0.812	5	0.00 (0/5)	Stable
27B	q4_K_M	0.914	[0.898, 0.929]	3	0.40 (2/5)	0.829	5	0.00 (0/5)	Stable
0.8B	q8_0	0.912	[0.912, 0.912]	1	0.80 (4/5)	—	0	1.00 (5/5)	Stable
2B	bf16	0.903	[0.892, 0.926]	3	0.40 (2/5)	0.760	2	0.60 (3/5)	Stable
27B	bf16	0.900	[0.886, 0.914]	5	0.00 (0/5)	0.855	5	0.00 (0/5)	Stable
4B	bf16	0.885	[0.873, 0.899]	5	0.00 (0/5)	0.741	5	0.00 (0/5)	Stable
2B	q8_0	0.876	[0.814, 0.931]	5	0.00 (0/5)	0.846	1	0.80 (4/5)	Stable
27B	q8_0	0.871	[0.833, 0.910]	5	0.00 (0/5)	0.839	5	0.00 (0/5)	Stable
4B	q4_K_M	0.861	[0.819, 0.899]	5	0.00 (0/5)	0.823	5	0.00 (0/5)	Stable
9B	q4_K_M	0.859	[0.820, 0.892]	5	0.00 (0/5)	0.830	5	0.00 (0/5)	Stable
9B	bf16	0.851	[0.811, 0.893]	5	0.00 (0/5)	0.815	5	0.00 (0/5)	Stable
35B MoE	bf16	0.846	[0.798, 0.887]	5	0.00 (0/5)	0.797	5	0.00 (0/5)	Stable
9B	q8_0	0.844	[0.783, 0.893]	5	0.00 (0/5)	0.795	5	0.00 (0/5)	Stable
35B MoE	q8_0	0.837	[0.815, 0.858]	5	0.00 (0/5)	0.803	5	0.00 (0/5)	Stable
4B	q8_0	0.824	[0.801, 0.846]	5	0.00 (0/5)	0.788	5	0.00 (0/5)	Stable
2B	q4_K_M	0.789	[0.612, 0.889]	3	0.40 (2/5)	0.772	4	0.20 (1/5)	Moderate
0.8B	bf16	—	—	0	1.00 (5/5)	—	0	1.00 (5/5)	Catastrophic

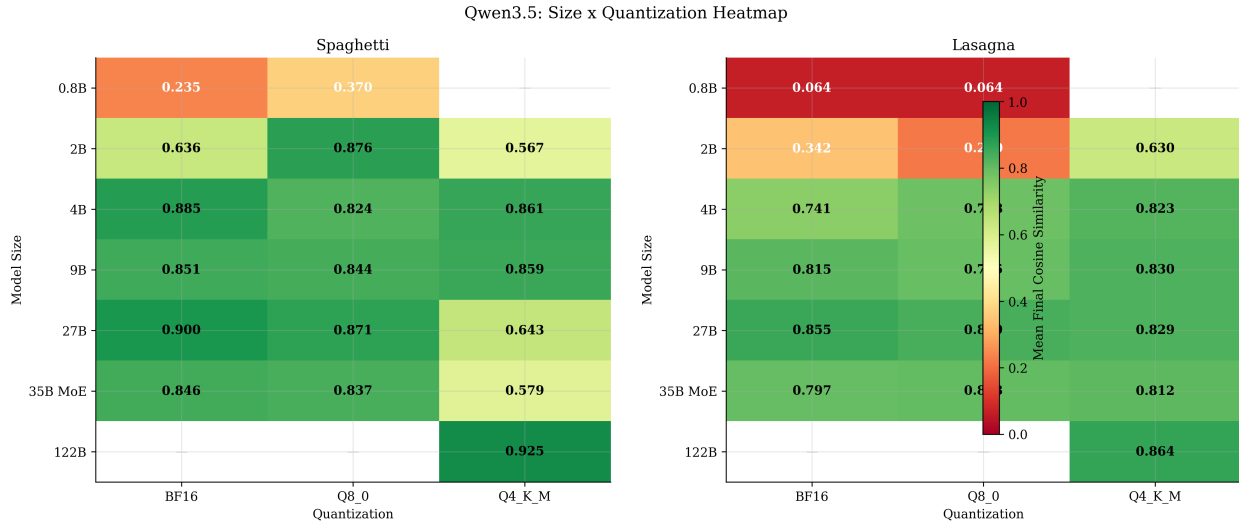


Figure 2: Mean final similarity by model size and quantization level. Left: spaghetti recipe. Right: lasagna recipe. Note that 0.8B Q4_K_M does not exist for Qwen3.5.

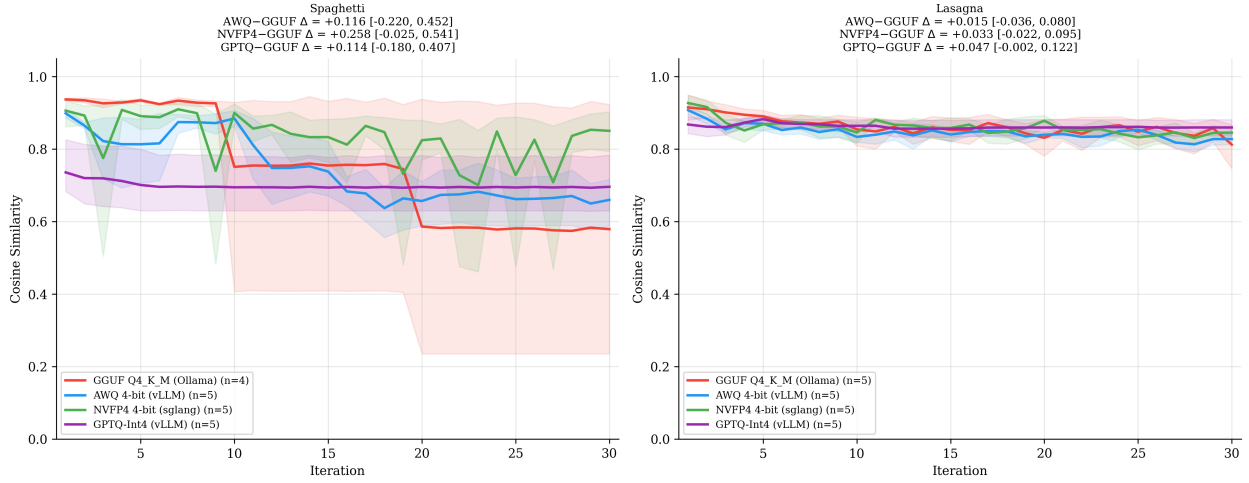


Figure 3: Four quantization methods for the 35B MoE model: GGUF Q4_K_M (Ollama), AWQ 4-bit (vLLM), NVFP4 4-bit (sglang), and GPTQ-Int4 (vLLM). Left: spaghetti. Right: lasagna. Panel titles give the seed-matched paired delta against GGUF with its bootstrap 95% CI; n per arm is shown in the legend.

4.2 RQ2: GGUF vs. AWQ vs. NVFP4 vs. GPTQ-Int4 Quantization

Figure 3 compares the 35B MoE model under four quantization methods: GGUF Q4_K_M (Ollama), AWQ 4-bit (vLLM), NVFP4 4-bit (sglang), and GPTQ-Int4 (vLLM). All four are 4-bit quantizations of the same underlying Qwen3.5-35B-A3B architecture, differing only in quantization algorithm and serving backend.

On the spaghetti recipe, GGUF Q4_K_M achieves 0.920 while AWQ drops to 0.660—a large delta of -0.260 . This suggests that the AWQ quantization method, despite being optimized for inference throughput, may introduce different error patterns than GGUF’s round-to-nearest approach. The difference is recipe-dependent: on the more complex lasagna prompt, AWQ achieves 0.828 while GGUF performance varies by quantization level.

NVFP4 (NVIDIA’s FP4 format) and GPTQ-Int4 provide two further data points, allowing us to distinguish between quantization algorithm effects and serving backend effects.

This finding extends Paper 1’s observation that AWQ and GGUF produce different drift dynamics for Qwen3-VL 30B, and suggests that the choice of quantization *method*—not just precision level—is a significant factor in multi-agent pipeline reliability.

4.3 RQ3: Temperature

Figure 4 shows temperature sensitivity across eight configurations (4 Ollama GGUF, the 35B AWQ, NVFP4 and GPTQ-Int4 variants, the 9B AWQ variant, and the Nemotron control).

Size-dependent temperature sensitivity. The 9B GGUF model shows remarkable temperature insensitivity: final similarity ranges only 0.823–0.890 across $T \in [0.0, 1.0]$ on spaghetti. In contrast, the 2B model shows *inverted* temperature sensitivity on spaghetti: $T = 0.0$ and $T = 0.3$ both sit at the empty-output floor of 0.235, and similarity then rises monotonically with temperature (0.512 at $T = 0.5$, 0.567 at $T = 0.7$, 0.691 at $T = 1.0$). The same inversion appears for the 35B MoE

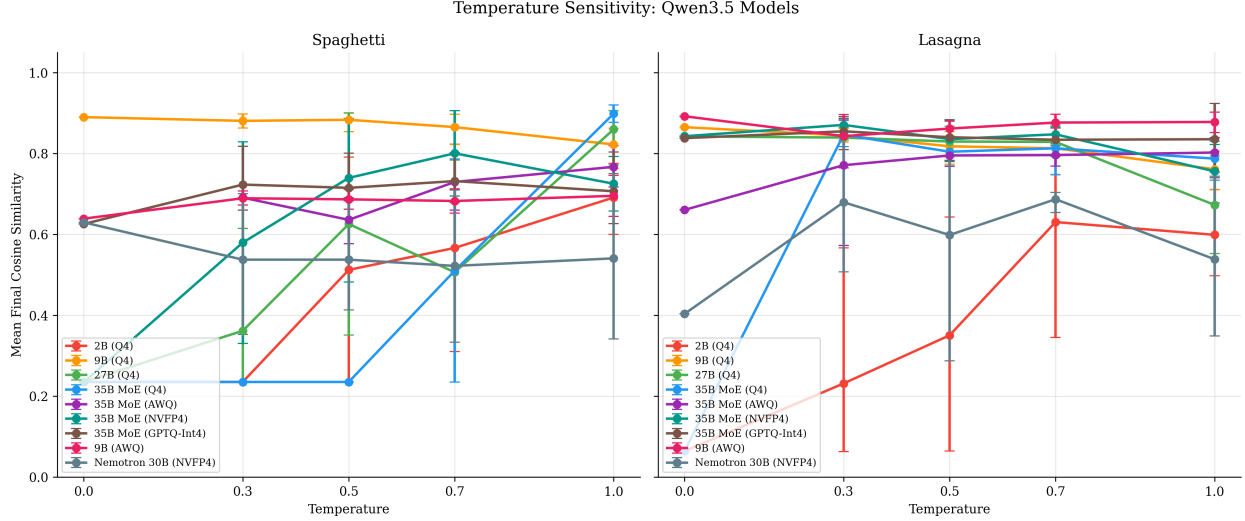


Figure 4: Temperature sensitivity across Qwen3.5 models. Left: spaghetti. Right: lasagna.

GGUF model, which is pinned at 0.235 for $T \leq 0.5$ and reaches 0.899 at $T = 1.0$. This suggests that low-temperature deterministic sampling can trap small models in repetitive attractor loops that the embedding model scores favorably, while moderate temperatures disrupt this lock-in without providing enough diversity to maintain content.

AWQ temperature profile. The 35B AWQ model shows a qualitatively different pattern: similarity *increases* with temperature on both recipes (spaghetti: 0.636 at $T = 0.5$ to 0.767 at $T = 1.0$; lasagna: 0.661 at $T = 0.0$ to 0.803 at $T = 1.0$; the AWQ $T = 0.0$ spaghetti chain was among those lost to the failed rerun). The 35B GPTQ-Int4 and 9B AWQ variants are by contrast nearly flat across the whole temperature range (GPTQ spaghetti 0.626–0.732; 9B AWQ spaghetti 0.639–0.696). This is the opposite of the conventional expectation that higher temperature increases randomness and thus drift. We hypothesize that the AWQ quantization creates a restricted output distribution at low temperatures, and higher temperature actually helps the model access more of its learned paraphrase space.

4.4 RQ4: System Prompts

We test whether Paper 1’s system prompt findings generalize within the Qwen3.5 family by running the same 20 system prompts across six Qwen3.5 configurations—9B and 27B GGUF (Ollama, Exp 4), 35B AWQ (Exp 3c), 35B NVFP4 (Exp 3d), 35B GPTQ-Int4 (Exp 3e) and 9B AWQ (Exp 3f)—plus the out-of-family Nemotron-3-Nano-30B control (Exp 3g). Each condition uses up to 5 completed chains on both the spaghetti and lasagna recipes; Table 3 summarizes the per-configuration spread, best and worst prompt, and the pairwise Spearman rank correlations, and Table 4 gives the per-prompt means.

Complexity threshold confirms. Consistent with Paper 1, system prompt effects are minimal on the spaghetti recipe but meaningful on lasagna across all three model sizes. The 35B AWQ model is a partial exception: it shows some system prompt sensitivity even on spaghetti, likely because AWQ’s lower baseline stability leaves more room for prompt engineering to help.

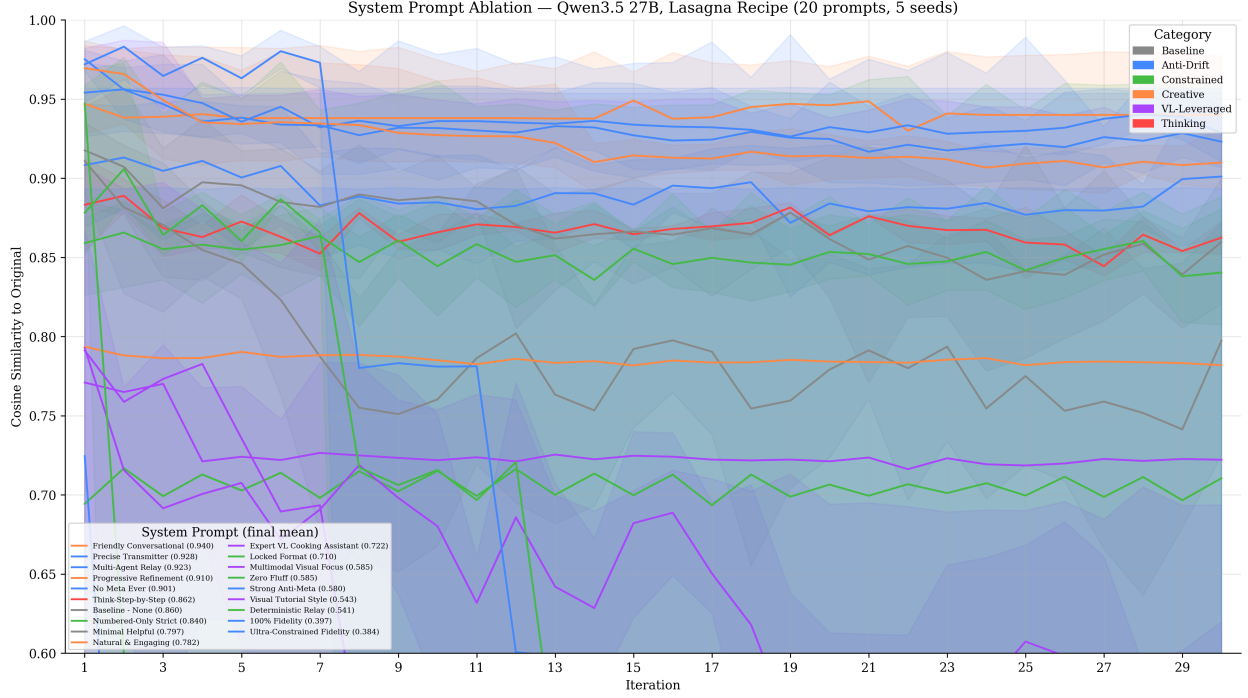


Figure 5: System prompt ablation on Qwen3.5 27B GGUF, lasagna recipe. 20 prompts shown with mean $\pm$ min/max band over the completed chains for each prompt. Prompts 15–20 rest on fewer than 5 chains and “Balanced Clarity” on none (Section 3).

Rankings are model-dependent. The best system prompts differ across model sizes (Figure 6). Prompts that work well for 27B may not transfer to 9B, confirming that prompt engineering must be tuned per model—even within the same family.

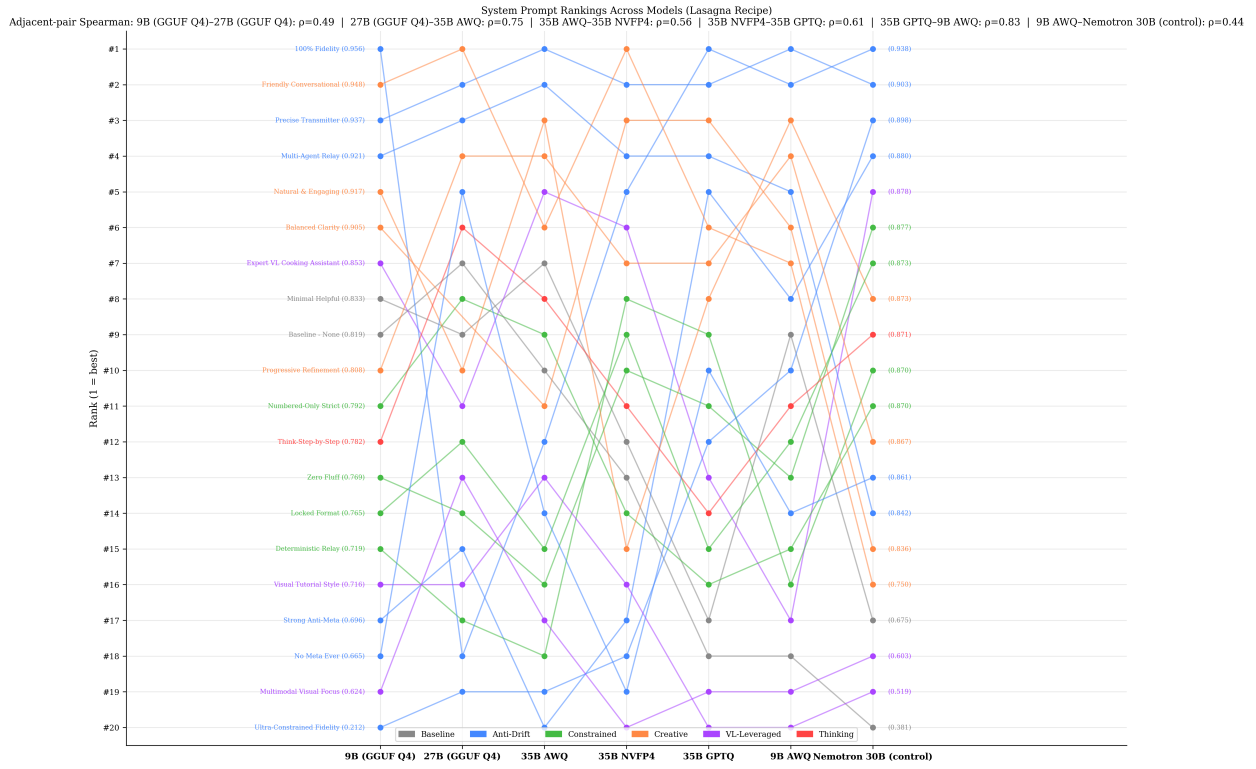


Figure 6: System prompt rankings across all six Qwen3.5 configurations and the Nemotron control (lasagna). Adjacent-column Spearman ρ values are annotated; the full matrix with bootstrap CIs is in Table 3.

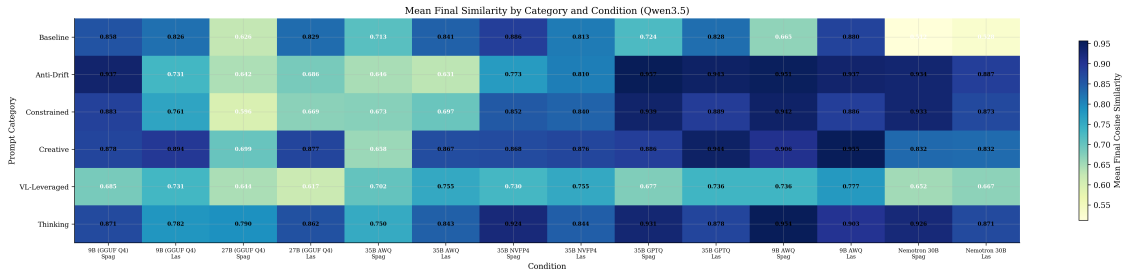


Figure 7: Mean final similarity by prompt category and condition.

Table 3: RQ4 summary (lasagna recipe), computed over *clean* chains only. Spread is the difference between the best- and worst-performing system prompt for that configuration; $n_{\min}$ is the smallest number of clean chains behind any prompt cell; “RA” is the configuration-wide runaway-thinking rate over all 20 prompts $\times$ 5 seeds.

Configuration	Mean	Spread	Best prompt	Worst prompt	$n_{\min}$	RA
9B (GGUF Q4)	0.822	0.332	100% Fidelity (0.956)	Multimodal Visual Focus (0.624)	1	0.05 (5/97)
27B (GGUF Q4)	0.868	0.380	Natural & Engaging (0.965)	Multimodal Visual Focus (0.585)	1	0.20 (16/81)
35B AWQ	0.742	0.743	Precise Transmitter (0.917)	Strong Anti-Meta (0.175)	5	0.00 (0/100)
35B NVFP4	0.854	0.294	Balanced Clarity (0.956)	Multimodal Visual Focus (0.663)	2	0.30 (30/99)
35B GPTQ	0.887	0.348	100% Fidelity (0.981)	Visual Tutorial Style (0.633)	4	0.00 (0/99)
9B AWQ	0.899	0.300	Precise Transmitter (0.972)	Visual Tutorial Style (0.672)	4	0.00 (0/99)
Nemotron 30B (control)	0.803	0.556	100% Fidelity (0.938)	Baseline - None (0.381)	5	0.00 (0/100)

Pairwise Spearman ρ between per-prompt mean rankings (20 prompts, lasagna), with 95% bootstrap CIs ($B = 10,000$).

Configuration A	Configuration B	ρ	95% CI
9B (GGUF Q4)	27B (GGUF Q4)	0.486	[-0.075, 0.897]
9B (GGUF Q4)	35B AWQ	0.779	[0.461, 0.918]
9B (GGUF Q4)	35B NVFP4	0.798	[0.444, 0.944]
9B (GGUF Q4)	35B GPTQ	0.549	[0.070, 0.825]
9B (GGUF Q4)	9B AWQ	0.633	[0.209, 0.857]
9B (GGUF Q4)	Nemotron 30B (control)	0.081	[-0.488, 0.573]
27B (GGUF Q4)	35B AWQ	0.746	[0.420, 0.897]
27B (GGUF Q4)	35B NVFP4	0.409	[-0.171, 0.819]
27B (GGUF Q4)	35B GPTQ	0.209	[-0.326, 0.707]
27B (GGUF Q4)	9B AWQ	0.312	[-0.202, 0.714]
27B (GGUF Q4)	Nemotron 30B (control)	-0.333	[-0.781, 0.204]
35B AWQ	35B NVFP4	0.556	[0.123, 0.800]
35B AWQ	35B GPTQ	0.259	[-0.200, 0.644]
35B AWQ	9B AWQ	0.480	[0.012, 0.794]
35B AWQ	Nemotron 30B (control)	-0.093	[-0.560, 0.390]
35B NVFP4	35B GPTQ	0.611	[0.180, 0.855]
35B NVFP4	9B AWQ	0.523	[0.080, 0.785]
35B NVFP4	Nemotron 30B (control)	0.162	[-0.369, 0.618]
35B GPTQ	9B AWQ	0.829	[0.542, 0.937]
35B GPTQ	Nemotron 30B (control)	0.496	[-0.060, 0.839]
9B AWQ	Nemotron 30B (control)	0.441	[-0.071, 0.784]

Table 4: System prompt effects on the lasagna recipe across all tested Qwen3.5 configurations plus the out-of-family Nemotron control. Mean final cosine similarity over the *clean* chains for that cell (chains containing no empty, runaway-thinking generation; up to 5 per cell). A superscript gives n_c when fewer than 5 clean chains survive; “—” marks a cell with no clean chain at all. The last two rows give the smallest per-cell n_c and the configuration-wide runaway-thinking rate. Sorted by the mean across configurations.

Prompt	Category	9B (GGUF Q4)	27B (GGUF Q4)	35B AWQ	35B NVFP4	35B GPTQ	9B AWQ	Nemotron 30B	Mean
Precise Transmitter	Anti-Drift	0.937	0.928	0.917	0.941	0.976	0.972 ⁴	0.903	0.939
Natural & Engaging	Creative	0.917	0.965 ⁴	0.914	0.929 ⁴	0.932	0.968	0.873	0.928
100% Fidelity	Anti-Drift	0.956 ⁴	0.917 ²	0.806	0.881 ⁴	0.981	0.970	0.938	0.921
Multi-Agent Relay	Anti-Drift	0.921	0.923 ³	0.915	0.917 ⁴	0.959	0.964	0.842	0.920
Balanced Clarity	Creative	0.905 ³	—	0.810	0.956 ³	0.969	0.959	0.836	0.906
Friendly Conversational	Creative	0.948	0.940 ¹	0.869	0.953 ⁴	0.940	0.924	0.750	0.903
Progressive Refinement	Creative	0.808	0.910	0.876	0.863	0.936	0.967	0.867	0.890
Expert VL Cooking Assistant	VL-Leveraged	0.853	0.890 ⁴	0.872	0.869	0.881	0.865	0.878	0.873
Think-Step-by-Step	Thinking	0.782	0.862 ²	0.843	0.844	0.878	0.903	0.871	0.855
Numbered-Only Strict	Constrained	0.792	0.840	0.842	0.811 ⁴	0.872	0.881	0.870	0.844
Zero Fluff	Constrained	0.769	0.941 ³	0.660	0.854 ³	0.903	0.893	0.877	0.842
Locked Format	Constrained	0.765	0.875 ⁴	0.745	0.846 ²	0.874	0.893	0.873	0.839
No Meta Ever	Anti-Drift	0.665	0.901 ²	0.746	0.841 ³	0.907	0.884	0.861	0.829
Minimal Helpful	Baseline	0.833	0.797	0.859	0.814	0.832	0.909	0.675	0.817
Deterministic Relay	Constrained	0.719	0.868 ³	0.542	0.861 ²	0.907	0.875	0.870	0.806
Strong Anti-Meta	Anti-Drift	0.857 ⁴	0.934 ³	0.175	—	0.943	0.924	0.880	0.786
Ultra-Constrained Fidelity	Anti-Drift	0.858 ¹	0.886 ²	0.228	—	0.892	0.907	0.898	0.778
Baseline - None	Baseline	0.819	0.860	0.822	0.813 ⁴	0.823	0.851	0.381	0.767
Visual Tutorial Style	VL-Leveraged	0.716	0.666 ⁴	0.766	0.710 ⁴	0.633	0.672	0.519	0.669
Multimodal Visual Focus	VL-Leveraged	0.624	0.585 ³	0.626	0.663 ³	0.694 ⁴	0.795	0.603	0.656
Minimum n_c		1	0	5	0	4	4	5	
Runaway rate		0.05	0.20	0.00	0.30	0.00	0.00	0.00	

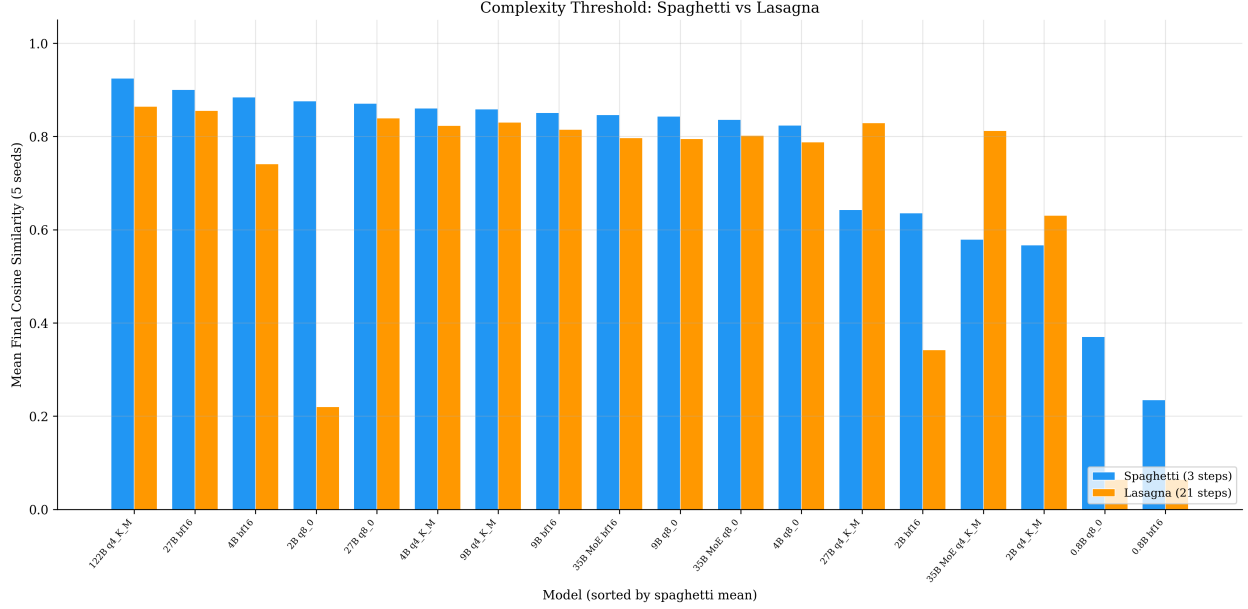


Figure 8: Spaghetti (3 steps) vs. lasagna (21 steps) for all Qwen3.5 models. Solid bars: spaghetti. Hatched bars: lasagna.

4.5 Complexity Threshold

Figure 8 compares spaghetti and lasagna performance across all 18 Ollama models.

Paper 1 found that system prompt effects only emerge with sufficiently complex prompts. The Qwen3.5 data confirms the underlying asymmetry between the two stimuli, and shows that it interacts with model size. Among the models that are stable on spaghetti, the lasagna mean is consistently the lower of the two but only modestly so (122B: 0.925 vs. 0.864; 27B BF16: 0.900 vs. 0.855; 9B Q4_K_M: 0.859 vs. 0.830). Small models show far larger gaps—2B Q8_0 drops from 0.876 on spaghetti to 0.220 on lasagna, and both 0.8B variants fall to 0.064—consistent with their struggling to carry a 21-step context through iterated generation. The three configurations whose spaghetti chains collapse into empty outputs (35B MoE Q4_K_M, 27B Q4_K_M, 2B Q4_K_M) invert the pattern and score *higher* on lasagna, which is a further indication that the short stimulus, not the long one, is what triggers the empty-output failure mode.

4.6 Cross-Generation Comparison

Figure 9 compares Qwen3 and Qwen3.5 at matched parameter counts using Q4_K_M quantization.

Dense model improvement at 27B. The 27B dense Qwen3.5 model is the only matched pair in which Qwen3.5 leads: 0.643 vs. Qwen3’s 32B dense at 0.564 (+0.079). Its bootstrap interval, [0.368, 0.918], comfortably contains the Qwen3 value, so this is a directional observation rather than a demonstrated improvement.

MoE models regress. The 35B MoE (Qwen3.5, 0.579) trails the 30B MoE (Qwen3, 0.806) by −0.227. Because the Qwen3.5 value is depressed by two chains that collapse to the empty-output floor, this gap reflects a difference in failure *rate* rather than in gradual fidelity loss.

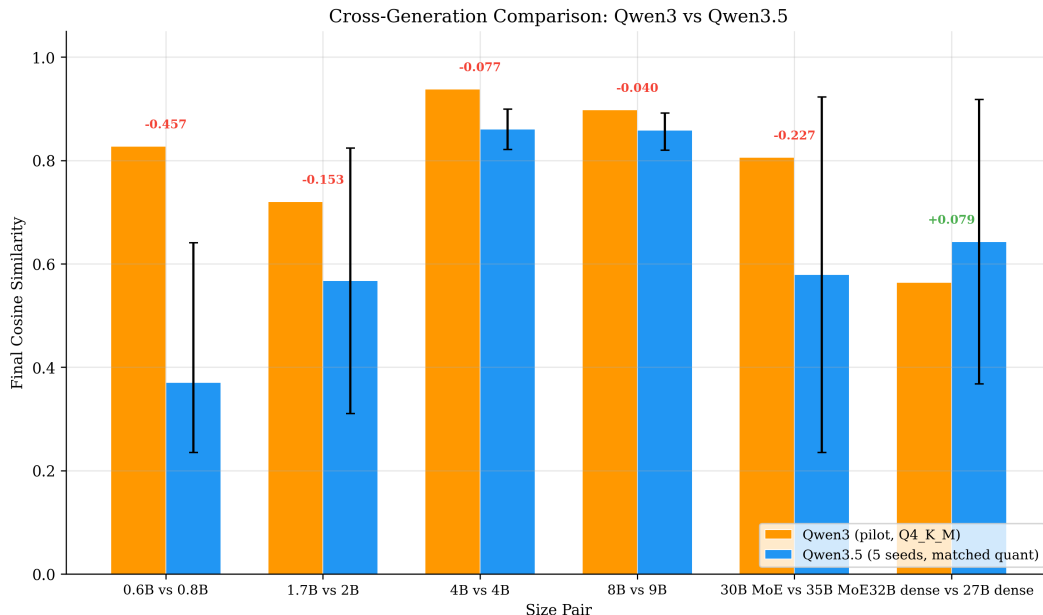


Figure 9: Cross-generation comparison: Qwen3 (pilot, single run) vs. Qwen3.5 (up to 5 completed chains, matched quantization). The Qwen3 bars carry *no* error bars because each is a single unreplicated run from the Paper 1 pilot; only the Qwen3.5 bars have bootstrap CIs.

Small model regression. Below 4B parameters, Qwen3.5 regresses compared to Qwen3: the 0.8B model (0.370) trails Qwen3’s 0.6B (0.827) by -0.457 , and the 2B (0.567) trails Qwen3’s 1.7B (0.720) by -0.153 . The 4B and 9B pairs are close to parity (-0.077 and -0.040). Taken together, five of six matched pairs favour Qwen3. We caution strongly that the Qwen3 arm comes from a single unreplicated pilot run with no confidence interval at all, while the Qwen3.5 means are averaged over up to five chains; the comparison should be read as motivating a properly seeded replication, not as evidence of a generational regression.

5 Discussion

5.1 Comparison with Paper 1

Paper 1 established a hierarchy of sensitivities: model choice $\gg$ infrastructure $>$ prompt engineering. The present within-family study reveals a more nuanced picture:

1. **Size matters, but not monotonically.** The stability threshold at 4B confirms that model capacity is necessary but not sufficient. Above this threshold, the ordering is irregular: 4B BF16 (0.885) outperforms 9B Q8_0 (0.844), and the 122B MoE (0.925) leads the best 35B MoE configuration (BF16, 0.846) by 0.079.
2. **Quantization effects are architecture-dependent.** Paper 1 found that “quantization is not monotonic.” We confirm and strengthen this: the direction and magnitude of quantization effects depend on model size and on the failure mode involved. Among the small dense models the reversal is between BF16 and Q8_0 (2B: 0.876 for Q8_0 vs. 0.636 for BF16); among the large models it is Q4_K_M that is anomalous, not because it degrades gradually but because it triggers whole-chain empty-output failures on the short stimulus.

3. **Quantization method matters.** GGUF, AWQ, NVFP4 and GPTQ-Int4, despite all being 4-bit methods, produce different drift dynamics for the same 35B MoE weights. This extends Paper 1’s Qwen3-VL finding to four quantization algorithms on the Qwen3.5 family and suggests that the quantization algorithm itself—not just the precision—shapes multi-step behavior, though at $n \leq 5$ chains per arm the individual pairwise intervals all include zero.
4. **Temperature sensitivity inverts at small scales.** Paper 1 found drift was “not merely a function of sampling entropy.” We find that for 2B models, the relationship between temperature and drift is actually *inverted* at low temperatures, suggesting qualitatively different dynamics below the stability threshold.

5.2 Practical Implications

For practitioners deploying Qwen3.5 in multi-agent pipelines:

- **Use ≥ 4 B models.** Models below 4B are unreliable for iterative tasks, regardless of quantization.
- **Watch for empty-output collapse under Q4_K_M.** On the short stimulus, the 27B and 35B MoE Q4_K_M configurations returned empty completions for the majority of iterations in several chains. Monitor for empty responses explicitly rather than relying on an aggregate similarity score.
- **Test your quantization method.** GGUF, AWQ, NVFP4 and GPTQ-Int4 are not interchangeable for drift-sensitive applications; benchmark all candidate formats on representative tasks.
- **Avoid low temperatures with small models.** $T = 0.3$ can produce worse drift than $T = 1.0$ for 2B models.

5.3 Limitations

1. **Single model family.** Results may not generalize to other architectures (Llama, Gemma, etc.).
2. **Paraphrase task only.** Other iterative tasks (summarization, translation) may show different patterns.
3. **Embedding metric.** Validation against bge-m3 shows strong agreement ($r = 0.825$, $\rho = 0.815$), but 9.4% of pairs were excluded due to bge-m3’s shorter context window (8K vs. 32K tokens). All excluded pairs are highly-drifted outputs (see Appendix D).
4. **Qwen3 comparison is unbalanced.** The Qwen3 data come from single pilot runs, while Qwen3.5 data have 5 seeds. The cross-generation comparison should be interpreted cautiously.
5. **Seeds are samples, not replicates.** Paper 3 of this series establishes that fixing the sampling seed does not pin a chain under batched inference on these serving stacks: identical seeds can produce different trajectories because batch composition and kernel scheduling vary between requests. The five “seeded” runs behind each cell should therefore be read as five independent samples from the model’s output distribution, not as reproducible replicates. All bootstrap intervals here are intervals over those samples.

6. **The lasagna stimulus is compressed.** The 21-step recipe actually sent to the models collapses steps 12–17 into the single line “Step 12–17: Layer sauce, noodles, ricotta, mozzarella (3 layers).” It is therefore a 16-line stimulus that nominally spans 21 steps (Appendix A). Results for the “21-step” condition should be read accordingly.
7. **Incomplete conditions.** Thirty-eight of the 2,007 chains did not reach iteration 30 and are excluded from every statistic (Section 3). Three of the affected serving endpoints have since been retired, so these conditions cannot be regenerated with the same software stack.
8. **Empty outputs and thinking overhead.** Several models (particularly 0.8B and 27B Q4_K_M) produce empty output strings in a substantial fraction of iterations. We attribute this to Qwen3.5’s think-before-responding architecture consuming the generation budget on internal reasoning tokens, but we cannot fully rule out serving-layer issues (e.g., Ollama silently truncating). The empty outputs are recorded as data and chains continue with the previous non-empty message, but this recovery mechanism may mask the true depth of drift.

6 Conclusion

We have presented the first controlled within-family analysis of semantic drift in iterated LLM paraphrase chains, studying the Qwen3.5 model family across 7 parameter sizes, 18 size/quant combinations, three serving backends (Ollama/GGUF, vLLM/AWQ and GPTQ-Int4, sglang/NVFP4), 5 temperatures, and 20 system prompts, in 1,969 completed 30-iteration chains totalling 59,070 inferences. Our key findings are:

1. A stability threshold at approximately 4B parameters, below which drift behavior becomes unpredictable.
2. Non-monotonic, size-dependent quantization effects, in which the ordering of BF16, Q8_0 and Q4_K_M reverses between model sizes.
3. Measurable differences between GGUF, AWQ, NVFP4 and GPTQ-Int4 quantization methods for the same underlying architecture, with the caveat that at $n \leq 5$ chains per arm the pairwise intervals overlap.
4. Inverted temperature sensitivity in small models, where low temperatures produce *worse* drift than high temperatures.
5. A cross-generation comparison in which five of six matched pairs favour Qwen3 over Qwen3.5, which we report as a call for a properly seeded replication given that the Qwen3 arm is a single pilot run.

These findings refine Paper 1’s sensitivity hierarchy: within a single model family, *size* dominates *quantization*, which dominates *temperature* and *prompt engineering*. However, the interactions between these factors are complex and non-monotonic, underscoring the need for empirical benchmarking of specific model configurations in drift-sensitive applications.

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A Prompt Texts

A.1 Spaghetti (3 steps)

Step 1: Boil water. Step 2: Add pasta for 8 minutes. Step 3: Drain and serve with sauce.

A.2 Lasagna (nominally 21 steps)

The following is the *verbatim* stimulus string sent to the models (PROMPTS["lasagna"] in `scripts/run_qwen35_size_quant.py`). Note that although the recipe is numbered through step 21, steps 12–17 are compressed into a single line in the stimulus itself, so the text actually transmitted contains 16 lines rather than 21. We reproduce it exactly rather than expanding it, because the compression is part of the condition being measured.

Follow these exact steps to prepare a classic homemade beef lasagna that serves 8-10 people: Step 1: Preheat your oven to 375°F (190°C) and lightly grease a 9×13 inch baking dish. Step 2: Bring a large pot of lightly salted water to a boil. Step 3: Heat 2 tablespoons of olive oil in a large skillet over medium heat. Step 4: Add 1 large finely diced onion and cook for 5 minutes until translucent. Step 5: Add 5 minced garlic cloves and cook for 1 minute until fragrant. Step 6: Add 1.25 lbs ground beef and 0.75 lb Italian sausage. Cook until browned. Step 7: Drain excess grease from the skillet. Step 8: Stir in two 28-oz cans of crushed tomatoes, 3 tbsp tomato paste, 2 tsp dried basil, 1.5 tsp dried oregano, 0.5 tsp fennel seed, 1 tsp sugar, 1 tsp salt, and 0.5 tsp black pepper. Step 9: Bring to a simmer and cook the meat sauce for 30 minutes. Step 10: Combine 15 oz ricotta, 1 egg, 1/2 cup Parmesan, 1/4 cup parsley, 1/2 tsp salt, 1/4 tsp pepper. Step 11: Cook 12 lasagna noodles until al dente. Drain and rinse. Step 12-17: Layer sauce, noodles, ricotta, mozzarella (3 layers). Step 18: Cover with foil and bake 25 minutes. Step 19: Remove foil and bake 25-30 more minutes until golden. Step 20: Rest 15-20 minutes before slicing. Step 21: Garnish with fresh basil and serve hot.

The spaghetti stimulus given above is likewise reproduced verbatim from PROMPTS["spaghetti"] in the same file.

B Model Inventory

All GGUF models were served with Ollama on a single on-premises GPU server. The vLLM models were `cyankiwi/Qwen3.5-35B-A3B-AWQ-4bit` (AWQ 4-bit), `Qwen/Qwen3.5-35B-A3B-GPTQ-Int4` (GPTQ-Int4) and `cyankiwi/Qwen3.5-9B-AWQ-4bit` (9B AWQ). The sglang models were `txn545/Qwen3.5-35B-A3B-NVFP4` and, as the out-of-family control, `nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-NVFP4`. Similarity was computed with Qwen3-Embedding-0.6B served by vLLM. Hardware and software versions are given in Appendix E.

Note: Qwen3.5 0.8B Q4_K_M does not exist as a released quantization. Qwen3.5 122B is only available in Q4_K_M due to GPU memory constraints. No official GPTQ-Int4 build of Qwen3.5 9B exists, and the available 9B NVFP4 builds come from publishers other than the one used for the 35B NVFP4 model, so a 9B NVFP4 arm was excluded to avoid a calibration confound.

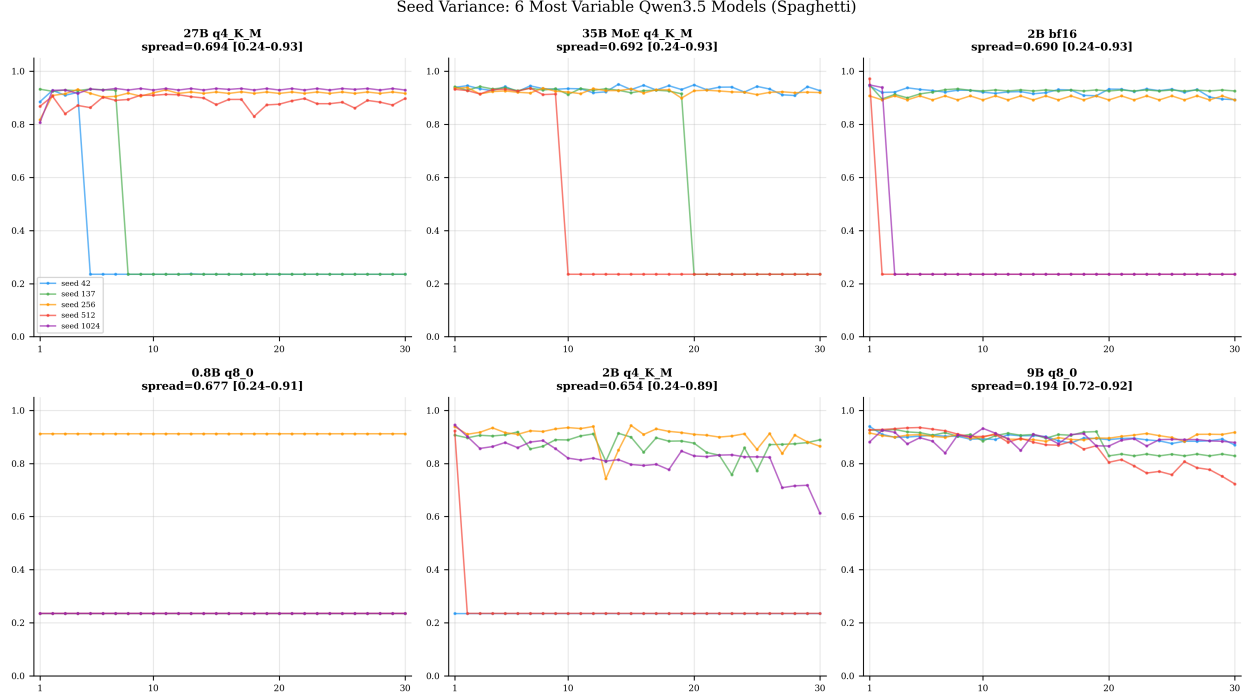


Figure 10: Seed variance for the 6 most variable Qwen3.5 models on the spaghetti recipe. Individual seed trajectories shown.

C Additional Figures

D Embedding Model Validation

Following Paper 1, we validate our primary embedding model (Qwen3-Embedding-0.6B, 32,768-token context) against bge-m3 (8,192-token context) by re-embedding all text pairs from the Qwen3.5 experiments and computing Pearson and Spearman correlations between the two models’ cosine similarity scores.

Of 22,167 total pairs, 20,072 (90.6%) produced valid similarities from both models, yielding Pearson $r = 0.825$ and Spearman $\rho = 0.815$ ($p < 10^{-10}$ for both). This strong agreement validates that our drift measurements are not artifacts of the specific embedding model chosen.

The remaining 2,095 pairs (9.4%) failed because bge-m3’s vLLM endpoint rejects inputs exceeding its 8,192-token context window rather than truncating them. All failed pairs are outputs exceeding 6,000 words, produced by chains that experienced runaway text expansion—a characteristic drift failure mode where certain system prompts (e.g., “Visual Tutorial Style,” “Friendly Conversational”) cause outputs to grow by several thousand words over 30 iterations. These excluded pairs have a mean Qwen similarity of 0.515 (median 0.572) compared to the dataset mean of 0.829, confirming they are systematically the most-drifted outputs. Because both embedding models would score these highly-diverged texts as low similarity, their exclusion is unlikely to meaningfully affect the correlation.

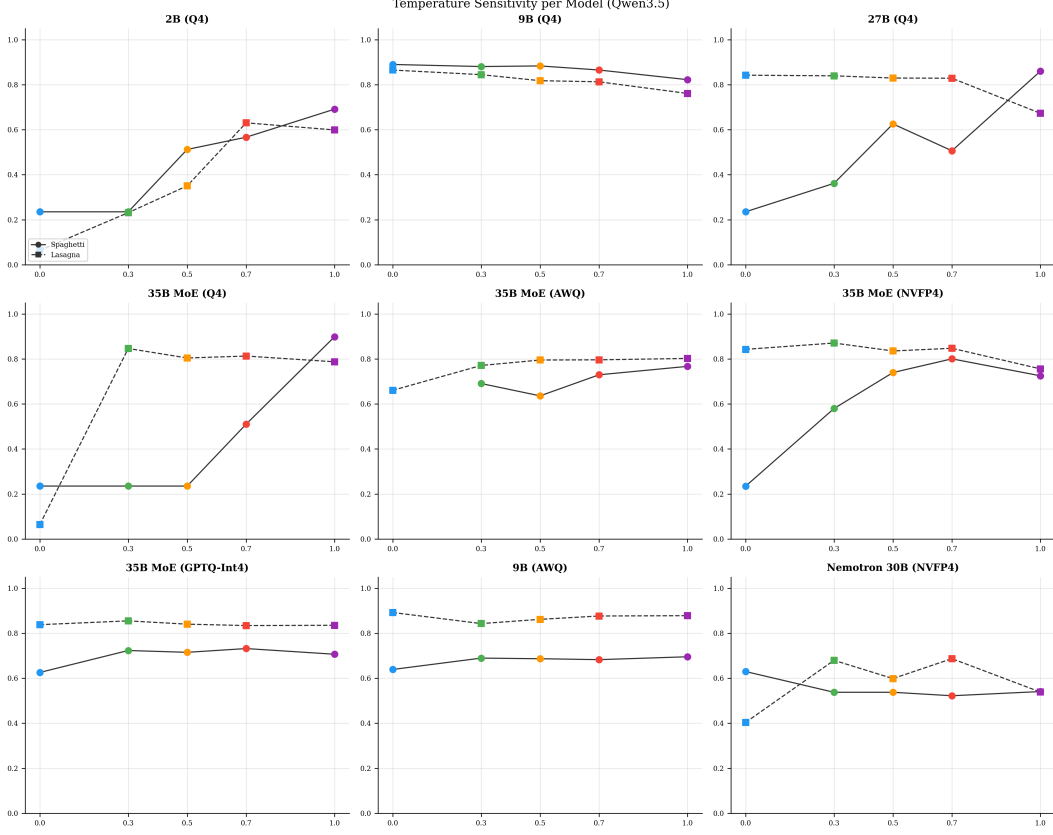


Figure 11: Temperature sensitivity per model: mean final similarity at each temperature for spaghetti (solid) and lasagna (dashed).

E Reproducibility and Compute

Hardware. All Ollama and vLLM runs were executed on a single on-premises server with one NVIDIA RTX PRO 6000 Blackwell GPU (96 GB). The sglang NVFP4 runs (35B NVFP4 and the Nemotron control) were executed on a second on-premises server with one NVIDIA RTX 5090 GPU (32 GB). Embeddings were computed with Qwen3-Embedding-0.6B served by vLLM on the same infrastructure.

Software. Ollama v0.16.3; vLLM v0.15.1, plus a CUDA 13.0 nightly build of vLLM for the GPTQ-Int4 and 9B AWQ runs; sglang for the NVFP4 runs. Analysis used Python with NumPy, SciPy and Matplotlib, driven by `scripts/analyze_qwen35_paper.py`; dependencies are pinned with `uv`.

Scale and compute budget. The study comprises 1,969 completed 30-iteration chains, that is 59,070 LLM inferences, drawn from 2,007 launched chains. Summing the recorded per-iteration wall-clock times across all Paper 2 data files gives approximately 630 hours of cumulative inference time; because several experiments ran with up to four concurrent workers, the elapsed wall-clock duration of the campaign was substantially shorter than that figure. Embedding validation added a further 22,167 embedding pairs (Appendix D).

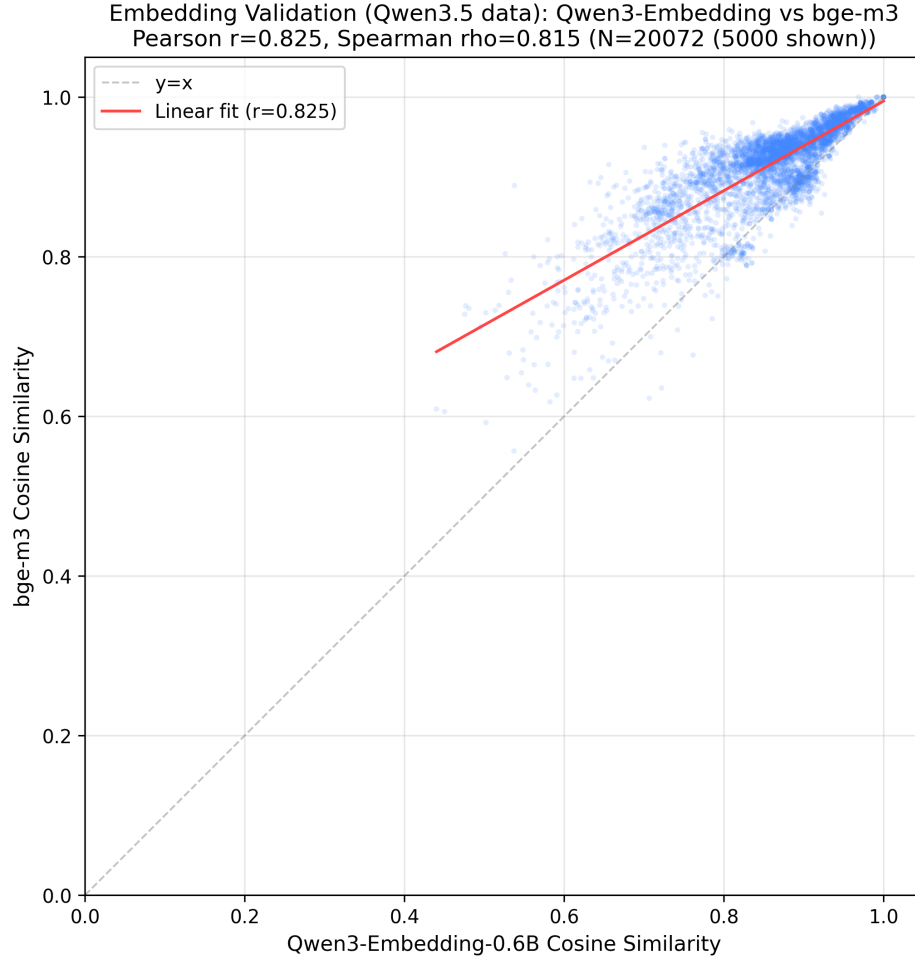


Figure 12: Embedding model validation scatter: Qwen3-Embedding-0.6B vs. bge-m3 cosine similarities for 20,072 Qwen3.5 experiment text pairs (2,095 pairs with outputs exceeding bge-m3’s 8,192-token context limit excluded). Pearson $r = 0.825$, Spearman $\rho = 0.815$.

Availability. The experiment scripts, raw per-iteration drift records and the analysis code that produces every figure and table in this paper are available from the author on request; a public repository will be linked here once one is released.